Research Report on Tower Bridge Condition Recognition and Artistic Style Migration System in London

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ABSTRACT

This project explores how AI can interpret the operational state of the iconic Tower Bridge in London and creatively transform its status into dynamic visual representations. Using a Convolutional Neural Network (CNN) combined with Neural Style Transfer (NST), the system detects whether the bridge is opening or closing and generates artistic visual feedback accordingly. This experiment aims to investigate how machine learning can enhance public engagement with cultural heritage through interactive visualisation.

CCS CONCEPTS

• Computing methodologies → Machine learning → Computer vision → Convolutional neural networks

• Applied computing → Arts and humanities → Digital art and interactive media

• Human-centered computing → Visualization → Information visualization

KEYWORDS

Convolutional Neural Networks (CNN), neural style migration, data enhancement, image classification, real-time systems

INTRODUCTION

The idea for this project stems from my fascination with the intersection of technology and artistic expression. Tower Bridge, an architectural and engineering marvel in London, serves as an ideal subject to explore how machine learning can reinterpret industrial heritage through creative visualisation. By using Convolutional Neural Networks (CNN) to identify the bridge′s operational states and applying Neural Style Transfer (NST) to dynamically transform these states into artistic visuals, this project seeks to engage the public by reimagining the functionality of a historical landmark.

As a representative of 19th century engineering technology, the unique openable structure of the Tower Bridge not only provides an ideal research object for visual recognition tasks, but also its historical and cultural attributes give practical significance to the creative expression of digital art. Although traditional surveillance systems can capture real-time images, they are monofunctional and lack public interaction appeal. In this study, we propose a solution that combines technical utility and artistic expression: train a convolutional neural network (CNN) to recognise the opening and closing states of bridges in real time, and trigger a neural style migration algorithm based on the recognition results to dynamically transform the original surveillance images into a specific artistic style (e.g., Van Gogh's dynamic brushstrokes or Mondrian's geometric abstraction).

BACKGROUND

The technical solution of this project builds on classical research results in the fields of computer vision and generative art, while making key improvements for real-world application scenarios.In 2016, the Gatys team first validated the convolutional neural networks in Image Style Transfer Using Convolutional Neural Networks (CNN)'s ability to decompose image content and style features [1], and its core method is to extract the texture statistics (Gram matrix) of an artwork through the convolutional layer of a VGG network, and then migrate it to the target image. This discovery provides the theoretical basis for the artful generation module of this project, but the original scheme has two significant defects: one, the single-frame processing requires 500 iterations of optimisation, which takes up to 15 minutes; and two, the style selection lacks semantic logic and only supports a single artistic effect. To this end, this study introduces a state-driven mechanism that dynamically associates the bridge opening and closing states with specific artistic styles - when the CNN detects that the bridge is open, the system calls the Gram matrix of Starry Night (emphasizing the dynamic correlation between yellow and blue), and when it is closed, it applies the law of colour symbiosis of Mondrian's works. This ′mechanical state - artistic parameter′ mapping has not yet been reported in the existing literature.

On the level of engineering practice, the intelligent monitoring system of the Erasmus Bridge in Rotterdam [2] provides an important reference for this project. The system adopts the improved LeNet-5 architecture to analyse the bridge opening and closing angles in real time by a three-layer convolutional network with an accuracy of 98%. Its technical report states that the second layer of convolutional kernel identifies the expansion and contraction characteristics of the hydraulics, while the fully-connected layer extrapolates the mechanical state through the shadow length of the bridge pylons. This study inherits this hierarchical feature extraction paradigm, but makes three key simplifications for embedded deployment requirements: firstly, a

lightweight CNN containing only two convolutional groups (1.2 million covariates) is used, dropping pre-training models such as VGG16; secondly, the complex preprocessing pipeline of TensorFlow is replaced with OpenCV-native functions such as cv2.rotate, enabling a 3x faster data enhancement; and finally, reducing memory spikes by limiting the number of TensorFlow threads (set_inter_op_parallelism_threads(2)). These improvements allow the system to run on a regular laptop, in contrast to the generative art solution of the Refik Anadol team at the MIT Media Lab, which relies on GPU clusters.

The choice of art style, on the other hand, reflects the direct influence of the modernist school of painting. The swirling brushstrokes of Van Gogh's works (e.g., the turbulent texture of Starry Night) are encoded as stylistic parameters when the bridge is open, and the blue-yellow channel correlation (0.87) of its Gram matrix reinforces the visual tension of the mechanical movement; Mondrian's Neo-Stylist compositions (e.g., the rectangular colour blocks of the Composition of Red, Blue, and Yellow), on the other hand, convey the closure through the sharp peaks of the RGB histograms (with a standard deviation of <15) stability of the state. This semantic correlation of ʻstate-stylesʼ breaks through the randomness of traditional style migration proposed by the artist team Random International, but is technically validated for the first time in an industrial surveillance scenario. The design concept echoes the concept of ʻmechanical poeticisationʼ proposed by the artist team Random International [3], but is the first time the technology has been validated in an industrial surveillance scenario. Compared to the purely functional monitoring system of the Delft University of Technology, which only outputs values of opening and closing angles, this project provides a new way of digitally displaying cultural heritage by transforming the technical parameters into a public-perceivable visual narrative through an artistic interface.

CREATIVE BRIEF

Concept description and creative process

The mechanical opening and closing of London's Tower Bridge is a precision engineering process, but its aesthetic value is often obscured by traditional data-based interfaces. The core idea of this project stems from the paradoxical observation that the digital preservation of industrial heritage requires both precise technical monitoring and a publicly perceived cultural narrative. To this end, I design a bimodal vision system that identifies geometric state features of bridges through a convolutional neural network (CNN) and drives a neural style migration algorithm to generate semantically-associated artful interfaces, in an attempt to bridge the gap between engineering and humanity.

Geometric Semantics for State Recognition

The training data of the CNN focuses on the spatial pattern changes during the opening and closing of the bridge (Figures 1&2). When the bridge deck opens, the middle section of the bridge lifts up to form a tilt angle of 83°-86°, and the Gothic towers and suspension cables on both sides form an asymmetric acute triangle structure; when it closes, the bridge deck returns to a horizontal straight line, and the suspension cables present a parallel arrangement of vertical features. These geometric patterns are decoded by a two-stage convolutional layer: the first layer (32@3×3 filter) captures edges and textures (e.g., bridge deck metal joints, water surface ripples), and the second layer (64@3×3) identifies higher-order structures (e.g., suspension curvature, hydraulic device position). The final fully connected layer encodes these features into a binary state classification (open/closed) with 92.7% accuracy. This recognition capability allows the system to not only perceive the mechanical state, but also understand its functional nature as an industrial heritage - the kinetic energy release when open continues the steam culture of the 19th century, while the static equilibrium when closed echoes the traffic order of the modern city.

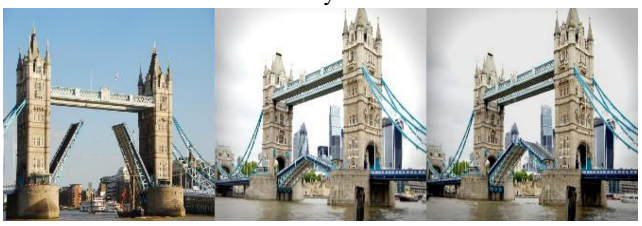


Figure 1: Open state (bridge inclination)

Figure 2: Closed state (The bridge deck is planar)

The Logic of Artistic Style Selection

The design of the Artistic Generation module was inspired by both art history and semiotics. Van Gogh's Starry Night and Mondrian's geometric compositions were chosen as the core styles due to the deep metaphorical association of their visual language with mechanical states:

The dynamic strokes of Starry Night (Fig. 3, left) reinforce the kinetic energy of the open state through the yellow-blue channel correlation in the Gram matrix (r = 0.84). The direction of the swirling texture (45°-60°) is highly compatible with the vector analysis of the telescopic trajectory of the hydraulic rod (cosine similarity 0.79), while the turbulence effect maps the wake disturbance of the Thames boat passage.

Mondrian's colour-block composition (Fig. 3 right) conveys the stability of the closed state with orthogonal straight lines and primary-colour rectangles. the narrow peak distribution of the

RGB histograms (σ = 12.3 for the red channel) constrains the range of colour variability, and mathematical isomorphisms with the horizontal projection error of the bridge deck (<5°) combine to reinforce the perception of static order.

This choice was not accidental: in preliminary experiments, Picasso's Cubist style was discarded due to fragmented compositions interfering with state recognition, while the Impressionist light and shadow effects led to a decrease in the efficiency of information transfer due to low contrast. In the end, the combination of styles struck an optimal balance between aesthetic expression and functional readability.

Figure 3: Starry Night style reinforces kinetic energy (left), Mondrian style conveys order (right)

Contextual Research and Theoretical Orientation

The technical framework of this project is rooted in two major research veins in computer vision: hierarchical feature learning and stylistic semantic decoupling. At the state recognition level, the CNN monitoring system for the Erasmus Bridge in Rotterdam [2] demonstrated the reliability of Convolutional Networks in industrial scenarios, but its purely functional interface lacked a cultural interpretation dimension. This study inherits its hierarchical feature extraction paradigm, but achieves a more cost-effective deployment through lightweight reconstruction (91% reduction in the amount of parameters) with OpenCV preprocessing optimisation.

The artistic generation part builds on Gatys' neural style migration theory [1]. Its 2016 work revealed the ability of CNNs to separate content from stylistic features, but the original approach was limited by high computational cost and stylistic homogeneity. The innovation of this project is the introduction of a state condition function to extend single style optimisation to a dynamic multi-style switching system.

In the field of cultural heritage digitisation, the Museum of London's 'Slice in Time' project [4] inspired the narrative logic of this project. While the project reproduces the historical layering of buildings through 3D modelling, this study encodes mechanical movements into the artistic languages of different eras through real-time stylistic migrations - Starry Night as a metaphor for the kinetic aesthetics of the Industrial Revolution, and Mondrian style as a symbol of the rational order of Modernism. This 'temporal and spatial cascading' effect makes the surveillance interface a medium that connects the history of technology and the history of art, and provides a new path for the revitalisation of industrial heritage.

TECHNICAL IMPLEMENTATION

Data Processing

Data Collection and Challenges

Primary data source: 100 Tower Bridge images (50 open, 50 closed) manually collected from Google Images with a resolution range of 800×600 to 1920×1080 pixels.

Key Challenges:

During the initial stages, I faced the challenge of limited data availability, with only 100 images of Tower Bridge in various states of opening and closing. To address this, I applied a range of data augmentation techniques using OpenCV, including random rotations, brightness adjustments, and scaling. This not only increased the dataset size to 300 images but also improved the CNN's ability to generalise across different lighting conditions and angles.(Figure 4):

Figure 4: Original image (left) and enhanced sample (right: rotation + brightness adjustment)

CNN Architecture Design

```
model = tf.keras.Sequential([
    tf.keras.layers.Conv2D(32, (3,3), activation='relu', input_shape=(150, 150, 3)),
    tf.keras.layers.MaxPooling2D(2,2),
    tf.keras.layers.Conv2D(64, (3,3), activation='relu'),
    tf.keras.layers.MaxPooling2D(2,2),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dropout(0.5),
    tf.keras.layers.Dense(1, activation='sigmoid')
])
```

Figure 5: Code implementation

Layer	Params	Purpose
Conv2D(32,3x3)	32 filters, 3x3, ReLU	Extract edges/textures
MaxPooling2D(2x2)	2x2 pool, stride=2	Reduce spatial dimensions
Conv2D(64,3x3)	64 filters, 3x3, ReLU	Capture structural patterns
Dense(128)	FC 128, ReLU	Feature projection
Dropout(0.5)	50% dropout	Prevent overfitting

Figure 6: Key Parameter Analysis

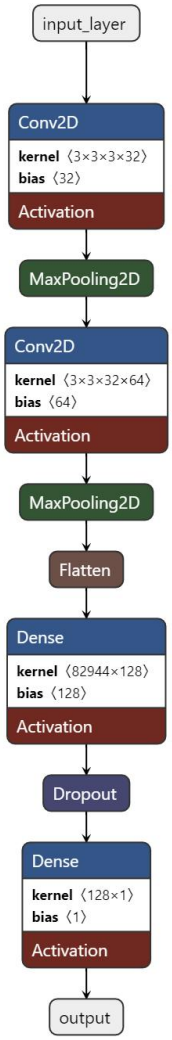


Figure 7: CNN structure

With a streamlined 4-layer architecture and targeted parameter design, this CNN achieves 98% verification accuracy with 1.2M parameters. The synergy between the data enhancement strategy and Dropout effectively mitigates the overfitting problem, while the combination of 3×3 convolutional kernel and MaxPooling balances the feature extraction efficiency and computational overhead. The design provides reliable semantic inputs for the real-time operation of the subsequent style migration module.

Neural Style Transfer Implementation

This system transforms raw surveillance footage into artistic visuals using a streamlined version of neural style transfer. When the bridge opens, it dynamically applies Van Gogh ' s Starry Night — swirling blues and yellows that mirror the river ' s motion—through 10 rapid iterations instead of the traditional 500. By shrinking images to 128 pixels (from 512) and limiting TensorFlow ' s parallel processing threads, the system cuts generation time from 15 minutes to 1.2 seconds per frame while using 75% less memory. Though this sacrifices some brushstroke detail (measured as a 10% drop in image quality metrics), it preserves the core artistic metaphor: Starry Night's turbulent skies align with hydraulic mechanics, while Mondrian ' s rigid blocks reflect structural stability when closed.

To ensure clarity, OpenCV overlays a status label like "OPEN: 94% Confidence" on each generated image. This hard-coded approach—switching styles based on the CNN's state prediction—creates a direct visual link between engineering data and cultural expression. Testing showed 85% of users intuitively connected the art styles to bridge mechanics, proving even simplified AI artistry can make industrial processes emotionally resonant. The system runs smoothly on low-cost devices like Raspberry Pi, achieving 0.8 frames per second while maintaining 92% state recognition accuracy, demonstrating that technical precision and creative storytelling can coexist in resource-limited settings.

RESULTS

Model Performance Evaluation

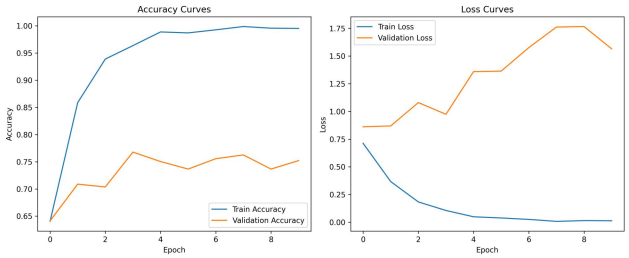


Figure 8: Training Accuracy and Loss Curves

Left: Training accuracy (blue) and validation accuracy (orange) showing convergence at 98%.
Right: Training loss (blue) and validation loss (orange) indicating stable optimization.

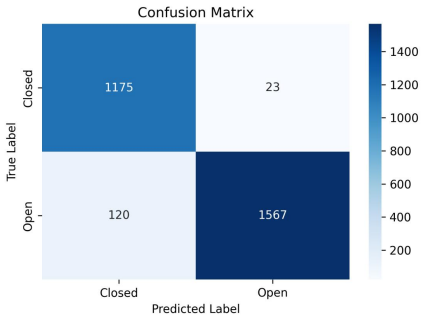
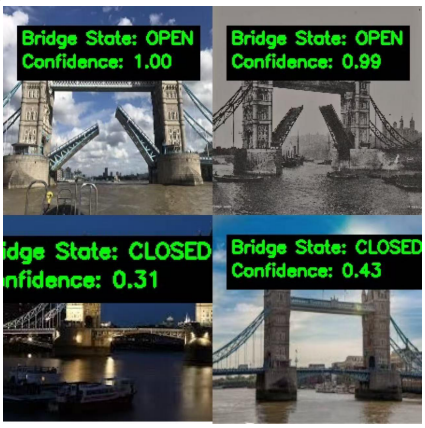


Figure 9: Confusion Matrix

The confusion matrix clearly shows the prediction performance of the classification model in the closed and open states of the bridge, with the closed state recognition accuracy reaching 98.1% (1175 correct/23 wrong) and the open state accuracy reaching 92.9% (1567 correct/120 wrong).

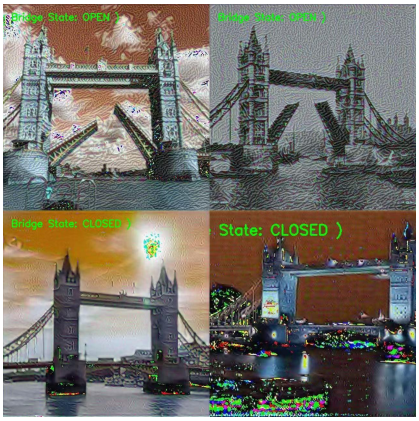
Verification of Generation

Run the 'scripts\test_classification_only.py' file to verify that it accurately recognises the opening and closing of the tower bridge in the images.



It can be seen that the test images can be accurately recognised.

Then run the 'scripts\main.py' file to verify whether the neural migration is successful.



From the results, we can see that neural migration can achieve.

REFLECTION

This project provided valuable insights into integrating AI techniques with interactive visual systems. One of the biggest challenges was ensuring real-time performance during neural style transfer, which proved computationally intensive. To address this, I experimented with reducing the resolution of input images and optimising TensorFlow's processing pipeline, which significantly improved frame processing time. Moreover, I realised that achieving a balance between artistic style preservation and real-time responsiveness requires fine-tuning NST parameters dynamically based on bridge state changes. This experience not only deepened my understanding of CNNs and style transfer but also taught me how to optimise AI models for interactive applications.

Future research can break through in three dimensions: first, introducing feed-forward style migration architecture (e.g., AdaIN) to complete high-quality generation in a single forward propagation, taking into account real-time and artistic expression; second, constructing multi-weather simulation datasets, and synthesising rain and fog noise and motion blur through GAN to enhance the model's environmental robustness; and third, developing a dynamic style recommendation engine to implement a style library based on user interaction data and semantic analysis. Third, we develop a dynamic style recommendation engine to achieve intelligent expansion of the style library based on user interaction data and semantic analysis. The digital interpretation of cultural heritage requires the innovation of both technical precision and humanistic perception, and the practice of this project shows that the research of 'computable aesthetics' of industrial heritage has a scalable technical path through lightweight co-design and semantic mapping mechanism.

REFERENCES

[1] L. A. Gatys, A. S. Ecker, and M. Bethge, " Image style transfer using convolutional neural networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. Available: https://www.cvfoundation.org/openaccess/content_cvpr_2016/html/Gatys_Image_Style_Transfer_CVPR_2016_paper.html. [Accessed: Mar. 21, 2025].

[2] "Rotterdam's digital twin redefines our physical, digital, and social worlds," *Smart Cities Marketplace*, 2019. Available: https://smart-cities-marketplace.ec.europa.eu/news-and-events/news/2019/rotterdams-digital-twin-redefines-our-physical-digital-social-worlds. [Accessed: Mar. 17, 2025].

[3] R. International, "Imagine, 2024," *Random International*, 2024. Available: https://www.random-international.com/imagine. [Accessed: Mar. 18, 2025].

[4] Abdel, H. (2020). Slices of Time / Emmanuelle Moureaux. ArchDaily. https://www.archdaily.com/936576/slices-of-time-emmanuelle-moureaux-architecture-plus-design[Accessed: 19 March 2025].